How have average new mortgage loan values changed over time across Canada, its provinces, and selected metropolitan areas between 2012 Q3 and 2025 Q4, and to what extent do time and geography explain differences in mortgage loan values?

Introduction

Mortgage lending is a useful indicator of housing market conditions because it reflects both the cost of purchasing property and the amount households are willing or able to borrow. Changes in average mortgage loan values can therefore capture broad shifts in affordability, housing demand, and regional market pressure. In Canada, these dynamics are especially important because mortgage markets differ sharply across provinces and metropolitan areas, with some jurisdictions experiencing much steeper price and borrowing growth than others. This makes mortgage values a useful empirical measure for studying both temporal change and geographic inequality in the housing market.

This report examines how average new mortgage loan values changed between 2012 Q3 and 2025 Q4 across Canada, the provinces, and a selected group of major metropolitan areas. The analysis combines descriptive statistics, graphical analysis, confidence intervals, hypothesis testing, bootstrapping, regression, and cross-validation in order to evaluate both overall trends and regional differences. The goal is not only to describe how mortgage values changed, but also to assess whether time and geography explain a substantial share of the observed variation.

Background of the data

This dataset records the average value of new mortgage loans for Canada, the provinces, and a set of census metropolitan areas over quarterly periods from 2012 Q3 to 2025 Q4. After cleaning and reshaping, the final dataset contained 2,430 observations, consisting of 54 quarterly observations for Canada**,** 540 province-level observations, and 1,836 CMA observations. The structure therefore supports both longitudinal analyses, by tracking change over time, and cross-sectional comparison, by examining how different places diverge from one another.

The final cleaned data include ten provinces and six selected CMAs used for closer metropolitan comparison: Halifax, Montréal, Ottawa, Toronto, Calgary, and Vancouver. The summary of the full dataset shows that average mortgage values ranged from $108,877 to $577,403, with an overall mean of about $264,144 and a median of about $246,774. This already suggests substantial heterogeneity across space and time, since the overall spread is very large relative to the mean.

Description of variables

The main response variable in this report is avg_mortgage_value, which measures the average dollar value of new mortgage loans for a given geography and quarter. This is the central variable used throughout the project because it captures how mortgage borrowing changes across both time and place. In substantive terms, higher values generally reflect more expensive housing markets or greater borrowing requirements, making this variable a useful indicator of regional mortgage market conditions.

The dataset also includes several time variables. quarter records the original reporting period, while year and quarter_num separate that label into a calendar year and quarter. quarter_date converts the quarter into a proper date format, which is useful for plotting trends over time. time_index gives each quarter a sequential numeric value, allowing time to be included directly in regression analysis and making it possible to estimate how mortgage values changed on average from one quarter to the next. Geography was handled through three related

variables. geo_level classifies each observation as Canada, Province, or CMA, which is important because these are different geographic scales and should not be treated as directly interchangeable. To make the analysis clearer, geography was then split into geo_province for Canada and provinces, and geo_cma for metropolitan areas. This separation made it possible to analyze provincial and metropolitan patterns independently, which improves both clarity and comparability in the results.

Research question

The main research question guiding this report is: How have average new mortgage loan values changed across Canada between 2012 Q3 and 2025 Q4, and how much of the variation in mortgage loan values can be explained by differences over time and across geography? More specifically, the report examines whether mortgage values followed a common national upward trend or whether provinces and selected metropolitan areas experienced distinct borrowing patterns and market trajectories of their own. This question is well suited to the dataset because it combines two central dimensions of the analysis: time, which captures how mortgage values changed quarter by quarter, and geography, which captures how those values differed across provinces and selected CMAs. It also provides a clear foundation for the regression analysis, since the model directly tests whether time and location explain a substantial share of the observed variation in average mortgage loan values.

Interpretation of tables and graphs

1. Structure of the Cleaned Full Dataset

Variable	Type	Example / Meaning
Rows		2,430 observations
Columns		10 variables
geography	<chr>	Geographic label such as "Canada"
geo_level	<chr>	Level of geography, e.g., Canada, Province, CMA
quarter	<chr>	Quarter code such as "2012Q3"
avg_mortgage_value	<dbl>	Average value of new mortgage loans
geo_province	<chr>	Province name when applicable
geo_cma	<chr>	CMA name when applicable; NA for non-CMA rows
year	<dbl>	Calendar year
quarter_num	<dbl>	Quarter number from 1 to 4
quarter_date	<date>	Quarter converted to date format
time_index	<int>	Sequential time counter used for trend analysis

The cleaned dataset contains 2,430 observations and 10 variables, which indicates that the mortgage data were successfully reshaped into a long format suitable for time series and geographic comparison. The structure shows that each row represents a specific geography-quarter combination, while the variables separate the geographic level, time information, and average mortgage loan value. The presence of year, quarter number, quarter date, and time index is especially useful because it allows the analysis to track changes over time in both descriptive graphs and regression models.

2. Summary Statistics for Average Mortgage Values

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

108,877	195,237	246,774	264,144	316,394	577,403

The summary statistics show substantial variation in average mortgage loan values across time and place. The minimum value is 108,877, while the maximum reaches 577,403, showing a very wide overall range. The mean of 264,144 is higher than the median of 246,774, which suggests that higher mortgage values in some locations or periods pull the average upward. This pattern points to a right-skewed distribution, meaning that while many observations are clustered at more moderate mortgage values, a smaller number of high-value observations increase the overall mean.

3. Number of Rows by Geography Level

Canada	CMA	Province
54	1,836	540

Most observations in the dataset come from census metropolitan areas, with 1,836 rows, followed by provinces with 540 rows, and Canada-level observations with 54 rows. This means the dataset is most detailed at the CMA level, which provides stronger coverage for metropolitan analysis than for national or provincial comparison. It also suggests that any patterns identified in the full dataset may be influenced heavily by metropolitan trends, since CMA observations make up the largest share of the data.

Table 4. Summary of Mortgage Values by Province

This table compares provincial average mortgage values and related descriptive statistics, sorted from the highest mean value to the lowest.

geo_province	mean_value	median_value	sd_value	min_value	max_value
British Columbia	395185	383977	64805	286652	500434
Ontario	346851	330302	77341	230770	475987
Alberta	309271	303610	22092	277158	361702
Canada	290997	281074	48690	212978	371063
Saskatchewan	233934	232304	14394	206354	270707
Manitoba	221081	215818	27450	177517	278687
Nova Scotia	204463	188444	39243	157664	285366
Newfoundland	203888	197600	17260	181331	253074
Prince Edward Island	193427	182226	43270	126763	266086
Québec	190498	183111	28173	153413	254499
New Brunswick	160792	145654	33210	122572	236329

British Columbia has the highest mean mortgage value at $395,185, followed by Ontario at $346,851, showing that the most expensive mortgage markets are concentrated in these provinces. Alberta and the overall Canada category follow next, while New Brunswick has the lowest mean value at $160,792. The spread of values also differs across provinces. Ontario has the largest standard deviation, which suggests greater internal variation, whereas Saskatchewan and Newfoundland show more stable values over time. Overall, the table indicates clear geographic inequality in mortgage loan values, with western and central provinces generally recording much higher averages than Atlantic provinces

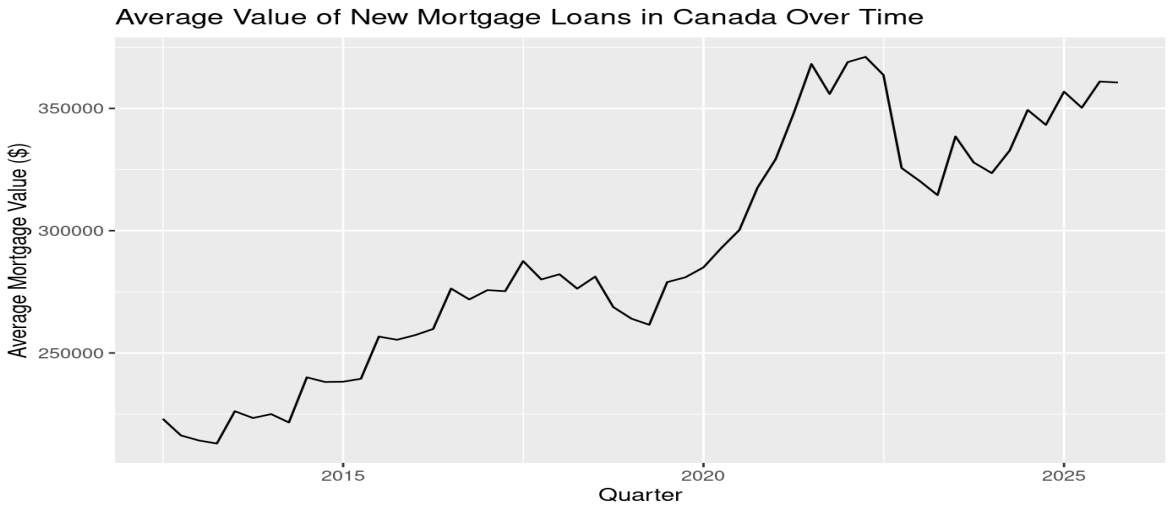
Table 5. Summary table for the 6 selected CMAs

This table compares average mortgage values across the selected metropolitan areas using the mean, median, standard deviation, minimum, and maximum values.

CMA	Mean Value	Median Value	SD Value	Minimum	Maximum
Vancouver	$456,449	$452,049	$70,220	$328,134	$577,067
Toronto	$427,126	$416,889	$87,327	$284,865	$577,403
Calgary	$331,998	$326,989	$26,067	$293,148	$393,044
Ottawa	$265,770	$250,764	$46,264	$204,723	$350,174
Montreal	$242,131	$227,297	$41,521	$187,490	$331,558
Halifax	$239,475	$220,741	$43,838	$187,748	$327,430

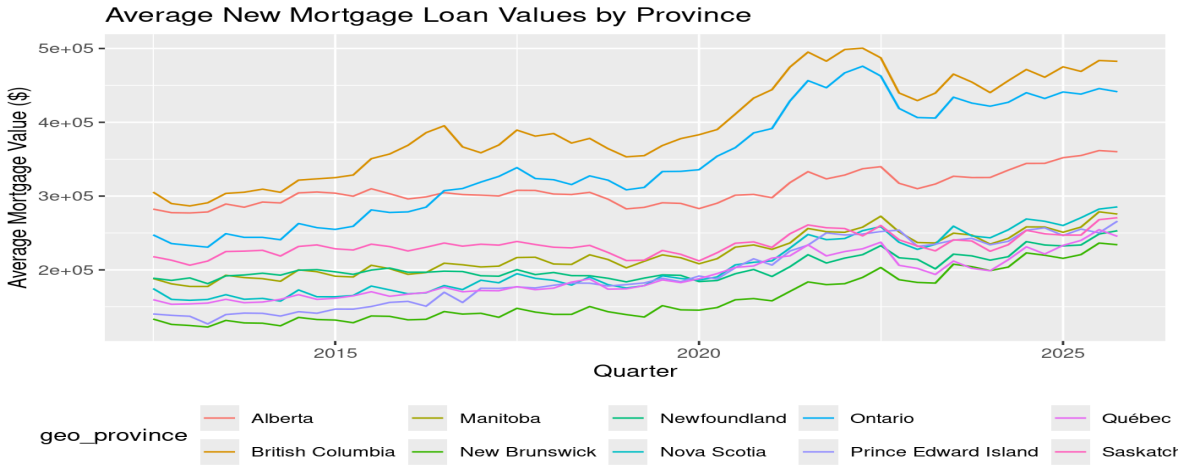
The selected CMAs show a clear hierarchy in mortgage loan values. Vancouver has the highest average mortgage value at $456,449, followed by Toronto at $427,126, which shows that these two markets are much more expensive than the others in the sample. Calgary sits in a middle position, while Ottawa, Montreal, and Halifax are considerably lower. Toronto also has the largest standard deviation, suggesting greater volatility across quarters, whereas Calgary is comparatively stable. Overall, the table shows that mortgage loan values differ substantially by metropolitan area, with Vancouver and Toronto consistently standing out as the most expensive markets.

Graph 1. Average Value of New Mortgage Loans in Canada Over Time



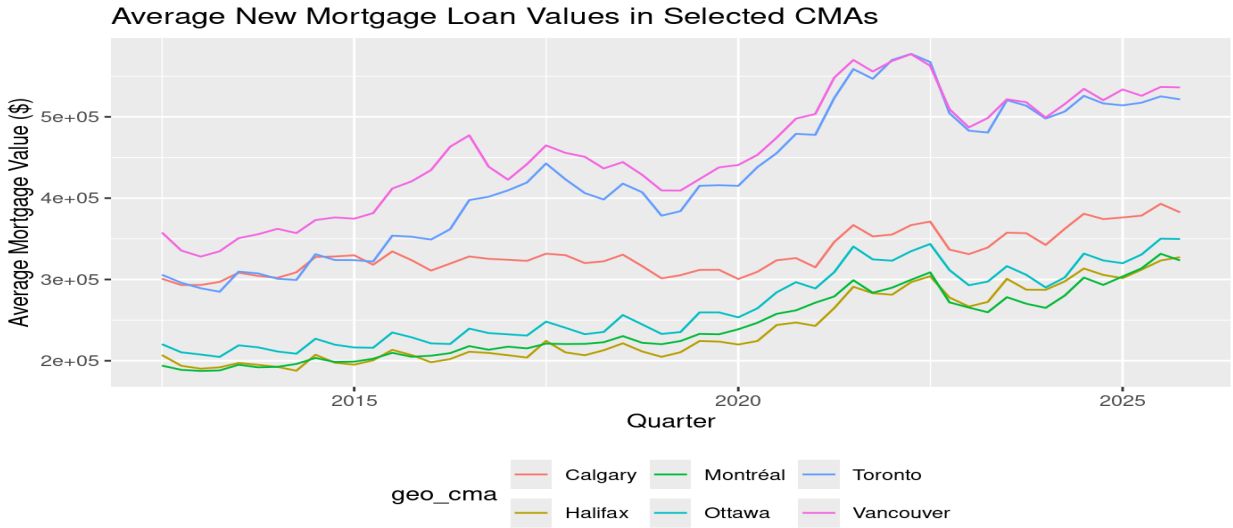
The graph shows a strong overall upward trend in average mortgage loan values in Canada from the early part of the series to 2025. Values rise steadily through the mid to late 2010s, then increase sharply around 2020 to 2022, where the series reaches its highest point. After that peak, there is a noticeable decline and some short-term fluctuation, but the values remain well above their earlier levels and start rising again toward the end of the period. This suggests that time plays an important role in explaining mortgage loan values, since the national pattern is not flat and instead shows substantial long-run growth with temporary corrections.

Graph 2. Trend in Average New Mortgage Loan Values Across Provinces (2012 Q3 to 2025 Q4)



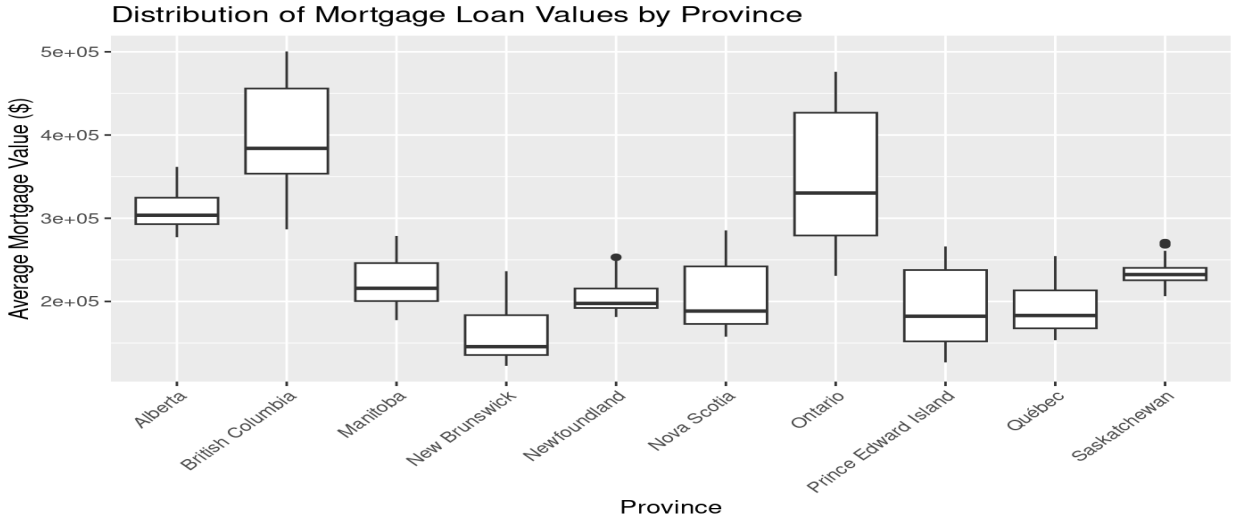
The provincial time-series graph shows that average new mortgage loan values increased over time in every province, but the levels and pace of change were very uneven. British Columbia stayed at the top for most of the period, with Ontario also consistently among the highest, and both provinces experienced especially strong growth after about 2020. Alberta remained in the upper-middle range and showed a steadier pattern, while provinces such as New Brunswick,
Prince Edward Island, Newfoundland, Nova Scotia, Manitoba, Québec, and Saskatchewan stayed at lower overall levels for much of the period. Even so, these lower-value provinces also trended upward, particularly in the later years. Overall, the graph suggests that time clearly matters, since mortgage values rose across Canada as the years passed, but geography also matters because some provinces, especially British Columbia and Ontario, remained far above others throughout the series.

Graph 3. Trend in Average New Mortgage Loan Values Across Selected CMAs (2012 Q3 to 2025 Q4)



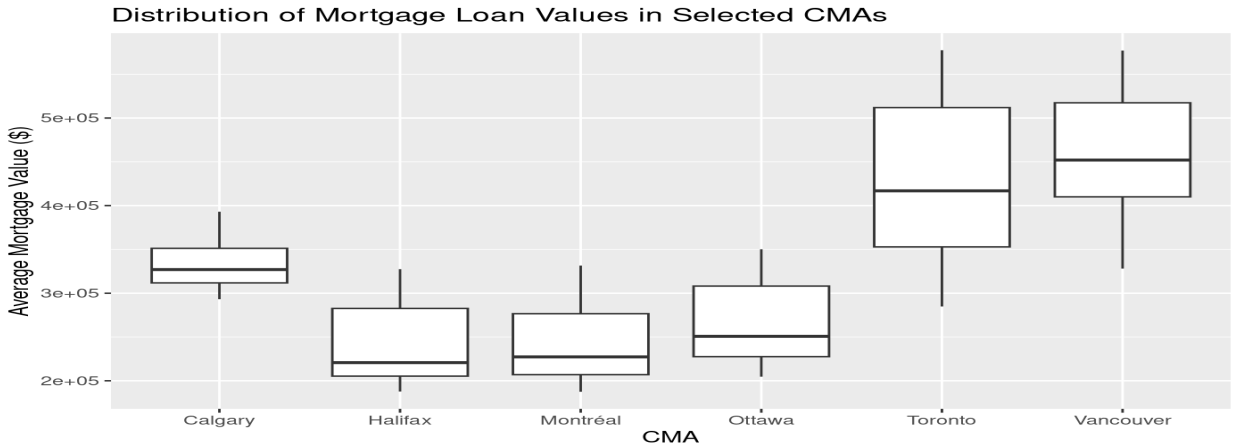
The selected CMA time-series graph shows the same broad upward pattern, but with even clearer differences between metropolitan areas. Vancouver and Toronto had the highest mortgage values across nearly the entire period, and both rose sharply, especially in the years after 2020, reaching peaks well above the other CMAs. Calgary formed a middle tier, while Ottawa, Montréal, and Halifax stayed lower for much of the timeline, though they also increased noticeably over time. The distance between Vancouver and Toronto and the lower-priced CMAs indicates that metropolitan housing markets were not moving at the same level even when they followed a similar direction. This means that rising mortgage values were not just a national time trend, but were also strongly shaped by local urban housing conditions.

Graph 4. Distribution of Average New Mortgage Loan Values by Province



The provincial boxplot highlights differences in the overall distribution of mortgage values within each province, not just their average levels. British Columbia and Ontario have the highest median values and also some of the widest spreads, which means they not only had more expensive mortgage markets overall, but also greater variation across time. Alberta appears relatively high as well, though less extreme than British Columbia and Ontario. In contrast, New Brunswick has the lowest median and a much lower overall range, while provinces such as Manitoba, Newfoundland, Nova Scotia, Québec, and Saskatchewan sit in the middle. Prince Edward Island shows a fairly wide spread relative to its median, suggesting noticeable variation over time even if its overall level is not among the highest. Taken together, this boxplot reinforces that provincial mortgage markets differed both in central tendency and in volatility.

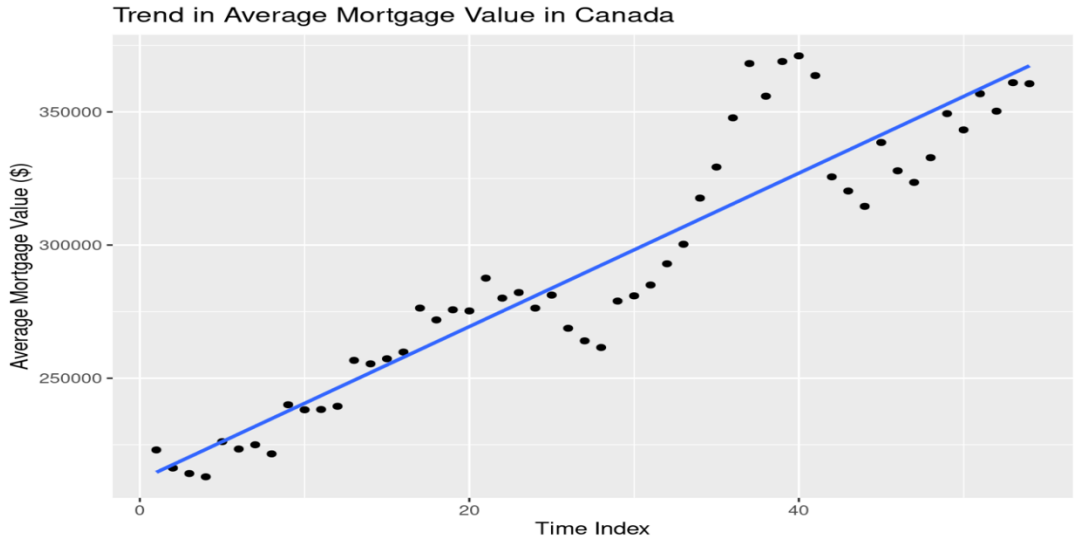
Graph 5. Distribution of Average New Mortgage Loan Values Across Selected CMAs



The CMA boxplot shows even stronger contrasts. Vancouver and Toronto stand out with the highest medians and the widest distributions, meaning they combined high typical mortgage values with substantial variation over time. Calgary is clearly below those two but still above Halifax, Montréal, and Ottawa in median terms. Ottawa's distribution appears broader than Montréal's and Halifax's, which suggests more fluctuation across the period, while Halifax and Montréal remain in the lower group. The spread for Vancouver and Toronto likely reflects the strong run-up in housing values over the study period(2012Q3 – 2025Q4) and the larger swings visible in the line graph.

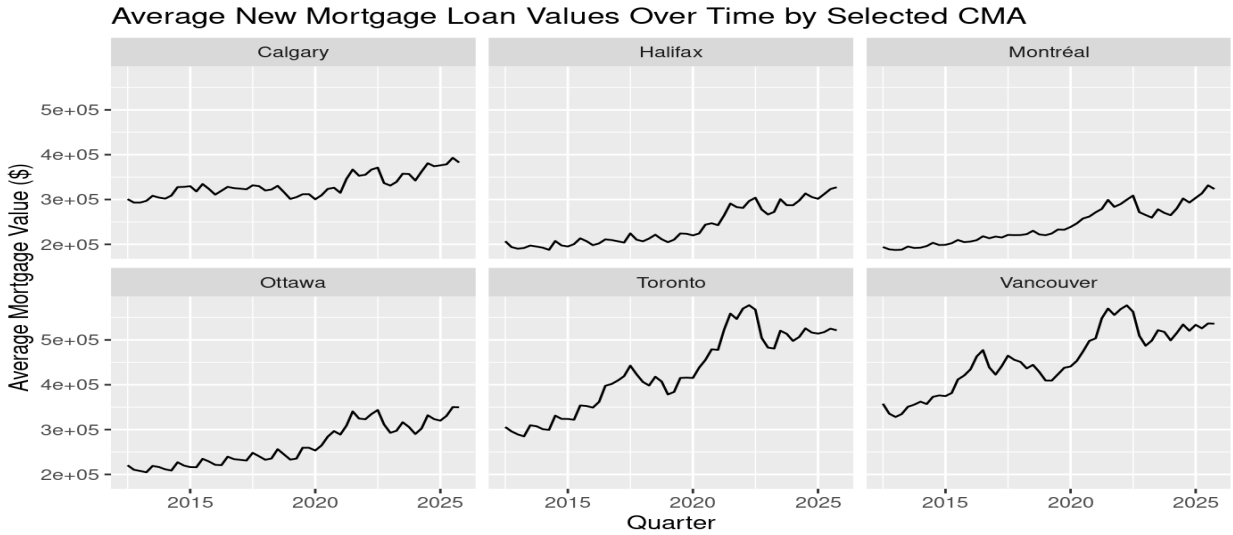
Overall, this boxplot confirms that the biggest metropolitan areas, especially Vancouver and Toronto, dominated the upper end of the mortgage value distribution, while the smaller CMAs remained considerably lower.

Graph 6. Linear Trend in Average Mortgage Loan Value in Canada



This scatterplot with the fitted regression line shows a clear positive relationship between time and average mortgage value in Canada. As the time index increases, mortgage values generally rise, which suggests that mortgage loan amounts increased steadily across the 2012 Q3 to 2025 Q4 period. Although there are some fluctuations around the line, the overall pattern is strongly upward.

Graph 7. Average New Mortgage Loan Values Over Time Across Selected CMAs



Inferential Analysis and Statistical Modelling of Mortgage Loan Values

1. Confidence Interval for Ontario Mean Mortgage Loan Value

One-sample t-test using Ontario observations only to estimate the mean average mortgage loan value and its 95% confidence interval.

Statistic	Value
Test type	One Sample t-test
Data	ontario_data$avg_mortgage_value
t statistic	32.956
Degrees of freedom	53
p-value	< 2.2e-16
Alternative hypothesis	True mean is not equal to 0

Measure	Lower / Value	Upper
95% confidence interval	325,740.7	367,960.8
Sample estimate	346,850.7	

The one-sample t-test estimates Ontario's average mortgage loan value at $346,850.7. The 95% confidence interval ranges from $325,740.7 to $367,960.8, which means the true average value for Ontario is likely to fall within that range. The p-value is extremely small, so the mean is statistically significant and therefore different from 0. Hence, the confidence interval tells us a plausible range for Ontario's average mortgage loan value across the quarters in the dataset.

2. Hypothesis Test: Ontario vs British Columbia

Welch two-sample t-test comparing the mean average mortgage loan value between Ontario and British Columbia across all quarters.

Statistic	Value
Test type	Welch Two Sample t-test
Data	avg_mortgage_value$geo_province
t statistic	3.5201
Degrees of freedom	102.85
p-value	0.0006443
Alternative hypothesis	Difference in means between British Columbia and Ontario is not equal to 0

Measure	Group / Lower	Value / Upper

95% confidence interval for mean difference	21,101.56	75,567.29
Mean in group British Columbia	British Columbia	395,185.2
Mean in group Ontario	Ontario	346,850.7

The Welch two-sample t-test shows that British Columbia and Ontario do not have the same average mortgage loan value. British Columbia has the higher sample mean at $395,185.2, compared with $346,850.7 for Ontario. Because the p-value is 0.0006443, the null hypothesis of equal means is rejected at the 5% level. The 95% confidence interval for the mean difference ranges from $21,101.56 to $75,567.29, suggesting that British Columbia's average mortgage loan value is likely between about $21,000 and $75,500 higher than Ontario's. Overall, this result supports the idea that geography matters, since two major provinces show statistically different average mortgage loan levels.

3. Bootstrap Results for Difference in Average Mortgage Loan Values

Summary table and interpretation

Observed Difference	Bootstrap Mean	Bootstrap SD	Lower 95% CI	Upper 95% CI
-48,334	-48,125	13,489	-73,997	-21,835

This table reports a bootstrap estimate of the difference in average mortgage loan values between two groups. The observed difference is -48,334, and the bootstrap mean is -48,125, which is very close to the original estimate. That similarity suggests the bootstrap resampling procedure produced a stable estimate of the difference. The bootstrap standard deviation of 13,489 shows the amount of sampling variability around that estimated difference.
The 95% bootstrap confidence interval ranges from -73,997 to -21,835. Because the entire interval is below zero, the result suggests that the true difference is likely negative rather than zero. In practical terms, this means the first group has a lower average mortgage loan value than the second group, and the difference appears statistically meaningful.

Regression Model 1 Output

Average mortgage value explained by time and province

Coefficient Estimates

Term	Estimate
Intercept	250,816
time_index	2,126
geo_provinceBritish Columbia	85,914
geo_provinceManitoba	-88,189
geo_provinceNew Brunswick	-148,479
geo_provinceNewfoundland	-105,383
geo_provinceNova Scotia	-104,807
geo_provinceOntario	37,580
geo_provincePrince Edward Island	-115,843
geo_provinceQuébec	-118,773

geo_provinceSaskatchewan	-75,337

Model Fit Statistics

Statistic	Value
Residual Standard Error	24,630
Degrees of Freedom	529
Multiple R-squared	0.9160
Adjusted R-squared	0.9144
F-statistic	576.7
Model p-value	< 2.2e-16

In this regression, Alberta is the baseline province because it is the one province not listed with its own coefficient. That means every province coefficient is interpreted relative to Alberta. For example, British Columbia's coefficient of 85,914 means its average mortgage loan value is estimated to be about $85,914 higher than Alberta's, holding time constant, while Ontario's coefficient of 37,580 means Ontario is about $37,580 higher than Alberta. Provinces with negative coefficients, such as Manitoba or Québec, are estimated to have lower average mortgage values than Alberta. The intercept of 250,816 is the model's predicted value for Alberta when time_index = 0, so it mainly serves as the model's starting point. The time_index coefficient of 2,126 means average mortgage values increase by about $2,126 per quarter, after accounting for provincial differences. Overall, the model fits very well: the R-squared of 0.916 means that 91.6% of the variation in mortgage loan values is explained by time and province together, and the very small p-value shows the model is statistically significant.

Interaction Regression Model Output and Summary

*Model 2: avg_mortgage_value ~ time_index * geo_province*

Table 1. Coefficients for the interaction model

Term	Estimate	Std. Error	t value	p-value
(Intercept)	278,106.6	4,263.9	65.224	< 2e-16
time_index	1,133.2	134.9	8.401	4.24e-16
geo_provinceBritish Columbia	12,692.1	6,030.0	2.105	0.03579
geo_provinceManitoba	-101,959.6	6,030.0	-16.909	< 2e-16
geo_provinceNew Brunswick	-171,093.6	6,030.0	-28.374	< 2e-16
geo_provinceNewfoundland	-97,334.9	6,030.0	-16.142	< 2e-16
geo_provinceNova Scotia	-137,839.2	6,030.0	-22.859	< 2e-16
geo_provinceOntario	-59,285.6	6,030.0	-9.832	< 2e-16
geo_provincePrince Edward Island	-157,975.0	6,030.0	-26.198	< 2e-16

geo_provinceQuébec	-132,825.1	6,030.0	-22.027	< 2e-16
geo_provinceSaskatchewan	-60,604.4	6,030.0	-10.050	< 2e-16
time_index:geo_provinceBritish Columbia	2,662.6	190.8	13.958	< 2e-16
time_index:geo_provinceManitoba	500.7	190.8	2.625	0.00892
time_index:geo_provinceNew Brunswick	822.4	190.8	4.311	1.95e-05
time_index:geo_provinceNewfoundland	-292.7	190.8	-1.534	0.12561
time_index:geo_provinceNova Scotia	1,201.2	190.8	6.296	6.48e-10
time_index:geo_provinceOntario	3,522.4	190.8	18.464	< 2e-16
time_index:geo_provincePrince Edward Island	1,532.1	190.8	8.031	6.50e-15
time_index:geo_provinceQuébec	511.0	190.8	2.679	0.00763
time_index:geo_provinceSaskatchewan	-535.7	190.8	-2.808	0.00517

Table 2. Overall model fit statistics

Statistic	Value
Residual standard error	15,450
Degrees of freedom	520
Multiple R-squared	0.9675
Adjusted R-squared	0.9663
F-statistic	814.7 on 19 and 520 DF
Model p-value	< 2.2e-16

Table 3. ANOVA comparison of Model 1 and Model 2

Model	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	529	3.2087e+11	—	—	—	—
2	520	1.2412e+11	9	1.9675e+11	91.59	< 2.2e-16

The interaction model allows each province to have its own starting level and its own rate of change over time. Alberta is the baseline province because it does not appear as a separate coefficient. The intercept of 278,106.6 is therefore the model's predicted mortgage value for Alberta when time_index equals 0, and the time_index coefficient of 1,133.2 represents Alberta's quarterly increase. The province coefficients show how each province differs from Alberta at the starting point, while the interaction terms show how each province's time trend differs from Alberta's slope.

Several interaction terms are statistically significant, which means provinces did not all follow the same growth path over time. For example, Ontario has a large positive interaction coefficient of 3,522.4, and British Columbia also has

a strong positive interaction coefficient of 2,662.6. This indicates faster growth in mortgage values over time in those provinces relative to Alberta. By contrast, Saskatchewan has a negative interaction coefficient of -535.7, which suggests a slower increase than Alberta, and Newfoundland's interaction term is not statistically significant, meaning its time trend is not clearly different from Alberta's.

The model fits the data extremely well. The multiple R-squared of 0.9675 means that about 96.75% of the variation in average mortgage loan values is explained by time, province, and their interaction. The adjusted R-squared of 0.9663 shows that the fit remains very strong even after accounting for the larger number of predictors. The overall F-statistic is highly significant, which confirms that the model as a whole explains mortgage values very well.

The ANOVA comparison shows that Model 2 performs much better than the simpler additive model. Adding the interaction terms reduces the residual sum of squares from 3.2087e+11 to 1.2412e+11, and the F-test of 91.59 with a p-value below 2.2e-16 shows that this improvement is statistically significant. In substantive terms, this means mortgage loan values not only changed over time and differed across provinces, but the rate of change over time also varied by province.

Cross-Validation and Supporting Inferential Tables

This section presents the 5-fold cross-validation results for Model 1, along with the Ontario confidence interval table and the Ontario versus British Columbia hypothesis test table.

Table 1. Five-Fold Cross-Validation Results for Model 1

Fold	RMSE ($)
1	22,139
2	26,000
3	23,764
4	25,343
5	26,205

Table 2. Mean RMSE Across All Five Folds

Mean RMSE ($)
24,690.17

The 5-fold cross-validation results show that Model 1 performs fairly consistently across different subsets of the data. The RMSE values range from 22,139 to 26,205, with a mean RMSE of 24,690.17. This means that when the model is used to predict mortgage loan values on unseen data, its predictions are off by about $24,690 on average. Because the fold-specific RMSE values are relatively close to one another, the model appears reasonably stable rather than performing well on only one part of the dataset. Overall, the cross-validation results suggest that the regression has good predictive ability, although some prediction error remains, which is expected in real housing data as it covers thousands of dollars.

Table 3. Ontario 95% Confidence Interval for Mean Mortgage Loan Value

Mean Estimate ($)	Lower 95% CI ($)	Upper 95% CI ($)
346,851	325,741	367,961

This confidence interval table estimates the mean mortgage loan value for Ontario across all quarters in the dataset. The sample mean is 346,851, and the 95% confidence interval ranges from 325,741 to 367,961. This suggests that the true average mortgage loan value for Ontario is likely to fall within that interval. The interval is fairly narrow relative to the scale of the values, which indicates a reasonably precise estimate of Ontario's average mortgage level.

Table 4. Welch Two-Sample t-Test: Ontario versus British Columbia

t Statistic	p Value	Lower 95% CI ($)	Upper 95% CI ($)
3.52	0.000644	21,102	75,567

The Welch two-sample t-test compares the mean mortgage loan values of Ontario and British Columbia. The p-value of 0.000644 is well below 0.05, so the null hypothesis of equal means is rejected. This indicates that the average mortgage loan value differs significantly between the two provinces. The 95% confidence interval for the difference runs from 21,102 to 75,567, which suggests that British Columbia's average mortgage value is higher than Ontario's by a substantial amount. Together, these results support the conclusion that geographic differences in mortgage loan values are statistically significant.

Appendix : R Code

Formatted code used for the mortgage loan values assignment

Data Cleaning and Preparation

```
##############################################
# STAA57 Project
# Average Value of New Mortgage Loans
# Data cleaning and preparation
##############################################

# readxl lets us read Excel files
library(readxl)

# tidyverse gives us dplyr, tidyr, ggplot2, stringr, etc.
library(tidyverse)

# lubridate helps us work with dates
library(lubridate)

##############################################
# 1. Import the Excel file
##############################################

# Read the Excel sheet exactly as it appears
# We use col_names = FALSE because the first few rows are title rows,
# not the actual variable names
```

```
mortgage_raw <- read_excel(
  "~/Statistical_Reports_Portfolio/average-value-new-mortgage-loans-ca-prov-cmas-2012-q3-
2025-q4-en.xlsx",
  sheet = "New mortgage loan",
  col_names = FALSE
)

##############################################
# 2. Extract the real column names
##############################################

# In this spreadsheet, row 4 contains the actual headers
# such as Geography, 2012Q3, 2012Q4, ..., 2025Q4
quarter_names <- mortgage_raw %>%
  slice(4) %>%
  unlist(use.names = FALSE) %>%
  as.character()

# Replace the default Excel column names (...1, ...2, etc.)
# with the actual names from row 4
colnames(mortgage_raw) <- quarter_names

##############################################
# 3. Keep only the rows that contain data
##############################################

# The first few rows of the spreadsheet contain title text and notes,
# not actual observations
mortgage_data <- mortgage_raw %>%
  slice(5:51) %>%
  rename(geography = Geography)

##############################################
# 4. Classify each row by geography type
##############################################

mortgage_data <- mortgage_data %>%
  mutate(
    geo_level = case_when(
      geography == "Canada" ~ "Canada",
      geography %in% c(
        "Newfoundland",
        "Prince Edward Island",
        "Nova Scotia",
        "New Brunswick",
        "Québec",
        "Ontario",
        "Manitoba",
        "Saskatchewan",
        "Alberta",
        "British Columbia"
      ) ~ "Province",
      geography %in% c("Provinces", "Selected Metropolitan Areas") ~ "header",
      TRUE ~ "CMA"
    )
  )

##############################################
# 5. Remove the header rows
##############################################

mortgage_data <- mortgage_data %>%
  filter(geo_level != "header")

##############################################
# 6. Convert the dataset from wide to long format
```

```
##############################################

mortgage_long <- mortgage_data %>%
  pivot_longer(
    cols = -c(geography, geo_level),
    names_to = "quarter",
    values_to = "avg_mortgage_value"
  )

##############################################
# 7. Clean the mortgage value column
##############################################

mortgage_long <- mortgage_long %>%
  mutate(
    avg_mortgage_value = gsub(",", "", avg_mortgage_value),
    avg_mortgage_value = as.numeric(avg_mortgage_value)
  )

##############################################
# 8. Create separate province and CMA variables
##############################################

mortgage_long <- mortgage_long %>%
  mutate(
    geo_province = case_when(
      geo_level == "Canada" ~ "Canada",
      geo_level == "Province" ~ geography,
      TRUE ~ NA_character_
    ),
    geo_cma = case_when(
      geo_level == "CMA" ~ geography,
      TRUE ~ NA_character_
    )
  )

##############################################
# 9. Create time variables
##############################################

mortgage_long <- mortgage_long %>%
  mutate(
    year = as.numeric(str_sub(quarter, 1, 4)),
    quarter_num = as.numeric(str_sub(quarter, 6, 6)),
    quarter_date = yq(str_replace(quarter, "Q", ": Q"))
  )

##############################################
# 10. Create a correct numeric time index
##############################################

quarter_lookup <- mortgage_long %>%
  distinct(quarter, year, quarter_num, quarter_date) %>%
  arrange(year, quarter_num) %>%
  mutate(time_index = row_number())

mortgage_long <- mortgage_long %>%
  left_join(
    quarter_lookup,
    by = c("quarter", "year", "quarter_num", "quarter_date")
  )

##############################################
# 11. Create separate province and CMA datasets
##############################################
```

```
mortgage_province <- mortgage_long %>%
  filter(geo_level %in% c("Canada", "Province"))

mortgage_cma <- mortgage_long %>%
  filter(geo_cma %in% c(
    "Toronto",
    "Vancouver",
    "Montréal",
    "Calgary",
    "Ottawa-Gatineau",
    "Halifax"
  )) %>%
  mutate(
    geo_cma = case_when(
      geo_cma == "Ottawa-Gatineau" ~ "Ottawa",
      TRUE ~ geo_cma
    )
  )

##############################################
# 12. Quick checks
##############################################

glimpse(mortgage_long)
summary(mortgage_long$avg_mortgage_value)
table(mortgage_long$geo_level)

mortgage_long %>%
  filter(geo_level == "Province") %>%
  distinct(geography)

unique(mortgage_cma$geo_cma)
```

Descriptive Statistics and Graphs

```
##############################################
# 13. Descriptive statistics tables
##############################################

mortgage_long %>%
  summarise(
    n = n(),
    mean_value = mean(avg_mortgage_value, na.rm = TRUE),
    median_value = median(avg_mortgage_value, na.rm = TRUE),
    sd_value = sd(avg_mortgage_value, na.rm = TRUE),
    min_value = min(avg_mortgage_value, na.rm = TRUE),
    max_value = max(avg_mortgage_value, na.rm = TRUE)
  )

province_summary <- mortgage_province %>%
  group_by(geo_province) %>%
  summarise(
    mean_value = mean(avg_mortgage_value, na.rm = TRUE),
    median_value = median(avg_mortgage_value, na.rm = TRUE),
    sd_value = sd(avg_mortgage_value, na.rm = TRUE),
    min_value = min(avg_mortgage_value, na.rm = TRUE),
    max_value = max(avg_mortgage_value, na.rm = TRUE)
  ) %>%
  arrange(desc(mean_value))

province_summary

cma_summary <- mortgage_cma %>%
```

```
  group_by(geo_cma) %>%
  summarise(
    mean_value = mean(avg_mortgage_value, na.rm = TRUE),
    median_value = median(avg_mortgage_value, na.rm = TRUE),
    sd_value = sd(avg_mortgage_value, na.rm = TRUE),
    min_value = min(avg_mortgage_value, na.rm = TRUE),
    max_value = max(avg_mortgage_value, na.rm = TRUE)
  ) %>%
  arrange(desc(mean_value))

cma_summary

##############################################
# 14. Graph 1: Canada over time
##############################################

ggplot(
  mortgage_province %>% filter(geo_province == "Canada"),
  aes(x = quarter_date, y = avg_mortgage_value)
) +
  geom_line() +
  ggtitle("Average Value of New Mortgage Loans in Canada Over Time") +
  xlab("Quarter") +
  ylab("Average Mortgage Value ($)")

##############################################
# 15. Graph 2: Provinces over time
##############################################

ggplot(
  mortgage_province %>% filter(geo_province != "Canada"),
  aes(x = quarter_date, y = avg_mortgage_value, color = geo_province)
) +
  geom_line() +
  ggtitle("Average New Mortgage Loan Values by Province") +
  xlab("Quarter") +
  ylab("Average Mortgage Value ($)") +
  theme(legend.position = "bottom")

##############################################
# 16. Graph 3: Selected CMAs over time
##############################################

ggplot(
  mortgage_cma,
  aes(x = quarter_date, y = avg_mortgage_value, color = geo_cma)
) +
  geom_line() +
  ggtitle("Average New Mortgage Loan Values in Selected CMAs") +
  xlab("Quarter") +
  ylab("Average Mortgage Value ($)") +
  theme(legend.position = "bottom")

##############################################
# 17. Graph 4: Province boxplot
##############################################

ggplot(
  mortgage_province %>% filter(geo_province != "Canada"),
  aes(x = geo_province, y = avg_mortgage_value)
) +
  geom_boxplot() +
  ggtitle("Distribution of Mortgage Loan Values by Province") +
  xlab("Province") +
  ylab("Average Mortgage Value ($)") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```
##############################################
# 18. Graph 5: CMA boxplot
##############################################

ggplot(
  mortgage_cma,
  aes(x = geo_cma, y = avg_mortgage_value)
) +
  geom_boxplot() +
  ggtitle("Distribution of Mortgage Loan Values in Selected CMAs") +
  xlab("CMA") +
  ylab("Average Mortgage Value ($)")

ggplot(
  mortgage_cma,
  aes(x = quarter_date, y = avg_mortgage_value)
) +
  geom_line() +
  facet_wrap(~ geo_cma) +
  ggtitle("Average New Mortgage Loan Values Over Time by Selected CMA") +
  xlab("Quarter") +
  ylab("Average Mortgage Value ($)")
```

Inferential Statistics and Modeling

```
##############################################
# Inferential statistics and modeling
##############################################

############################
# 1. Confidence Interval
############################

ontario_data <- mortgage_province %>%
  filter(geo_province == "Ontario")

ontario_ci <- t.test(ontario_data$avg_mortgage_value, conf.level = 0.95)
ontario_ci

############################
# 2. Hypothesis Test
############################

ontario_british_columbia_data <- mortgage_province %>%
  filter(geo_province %in% c("Ontario", "British Columbia"))

province_test <- t.test(
  avg_mortgage_value ~ geo_province,
  data = ontario_british_columbia_data,
  conf.level = 0.95
)

province_test

############################
# 3. Bootstrapping
############################

set.seed(57)

ontario_values <- ontario_british_columbia_data %>%
  filter(geo_province == "Ontario") %>%
```

```
  pull(avg_mortgage_value)

bc_values <- ontario_british_columbia_data %>%
  filter(geo_province == "British Columbia") %>%
  pull(avg_mortgage_value)

observed_diff <- mean(ontario_values) - mean(bc_values)
B <- 5000
boot_diffs <- numeric(B)

for (i in 1:B) {
  boot_ontario <- sample(ontario_values,
                         size = length(ontario_values),
                         replace = TRUE)
  boot_bc <- sample(bc_values,
                    size = length(bc_values),
                    replace = TRUE)
  boot_diffs[i] <- mean(boot_ontario) - mean(boot_bc)
}

bootstrap_summary <- tibble(
  observed_difference = observed_diff,
  bootstrap_mean = mean(boot_diffs),
  bootstrap_sd = sd(boot_diffs),
  lower_95_CI = quantile(boot_diffs, 0.025),
  upper_95_CI = quantile(boot_diffs, 0.975)
)

bootstrap_summary

ggplot(tibble(boot_diffs = boot_diffs), aes(x = boot_diffs)) +
  geom_histogram(bins = 30) +
  ggtitle("Bootstrap Distribution of Mean Difference: Ontario - British Columbia") +
  xlab("Difference in Mean Mortgage Value ($)") +
  ylab("Frequency")

############################
# 4. Regression Analysis
############################

province_reg_data <- mortgage_province %>%
  filter(geo_province != "Canada") %>%
  mutate(
    geo_province = as.factor(geo_province)
  )

model_1 <- lm(avg_mortgage_value ~ time_index + geo_province,
              data = province_reg_data)

summary(model_1)

ggplot(province_reg_data, aes(x = time_index, y = avg_mortgage_value)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE) +
  ggtitle("Scatterplot with Linear Trend: Mortgage Value vs Time") +
  xlab("Time Index") +
  ylab("Average Mortgage Value ($)")

model_2 <- lm(avg_mortgage_value ~ time_index * geo_province,
              data = province_reg_data)

summary(model_2)
anova(model_1, model_2)

############################
# 5. Cross-Validation (5-fold cross-validation)
```

```
############################

set.seed(57)
k <- 5

province_reg_data <- province_reg_data %>%
  mutate(fold = sample(rep(1:k, length.out = n())))

rmse_values <- numeric(k)

for (j in 1:k) {
  train_data <- province_reg_data %>%
    filter(fold != j)

  test_data <- province_reg_data %>%
    filter(fold == j)

  cv_model <- lm(avg_mortgage_value ~ time_index + geo_province,
                 data = train_data)

  predictions <- predict(cv_model, newdata = test_data)

  rmse_values[j] <- sqrt(mean((test_data$avg_mortgage_value - predictions)^2))
}

cv_summary <- tibble(
  fold = 1:k,
  RMSE = rmse_values
)

cv_summary
mean_rmse <- mean(rmse_values)
mean_rmse

############################
# 6. Simple tables for report writing
############################

ci_table <- tibble(
  mean_estimate = ontario_ci$estimate,
  lower_95_CI = ontario_ci$conf.int[1],
  upper_95_CI = ontario_ci$conf.int[2]
)

ci_table

test_table <- tibble(
  t_statistic = province_test$statistic,
  p_value = province_test$p.value,
  lower_95_CI = province_test$conf.int[1],
  upper_95_CI = province_test$conf.int[2]
)

test_table

regression_table <- summary(model_1)$coefficients
regression_table
```